Vibrotactile Fidget Band for Combating Hyperactivity

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Figure [1]: Vibrotactile fidget band designed to replace the fidgeting methods like clicking a pen and drumming fingers on a table, as seen to the left.

***Abstract*— Many people today struggle with a small attention span, which hinders the ability to focus on school, work, or everyday tasks. The goal of this paper is to create a haptic device to be worn on the wrist that mimics intuitive methods of fidgeting, such as tapping a finger on a desk, clicking a pen, or cracking knuckles, through a variety of vibrations to increase focus and concentration. Participants will complete PEBL CPT, which is a commonly used low-cognitively demanding test, with and without the device. Their score and responses to a questionnaire given after the exams will be used to determine if the device has an effect on performance, self-reported concentration levels, and observed fidgeting frequency. These findings will determine if haptic devices can serve as a tool to support focus.**

I. INTRODUCTION

Many people have difficulty maintaining focus during everyday tasks in a work or school environment. To combat losing focus, it is common for people to fidget by drumming fingers on a desk, clicking a pen, or cracking knuckles. This paper focuses on creating a haptic device to be worn on the wrist that mimics intuitive methods of fidgeting previously mentioned, through vibration patterns.

In current research, most devices created to combat fidgeting require some attention to the device and very often encumbers the user's hands in some way [1,2,3]. This is not ideal because most often these devices are used in a school or work environment, which typically requires the use of both hands. Other devices have been created to implement technology to discourage fidgeting in commonly used tools, such as a pen [4]. However, this limits the device to only be used when using a pen, which is a significant disadvantage for those who, instead of using a pen, work on computers for extended periods of time, for example. In this paper, we aim to create a wrist-worn device to produce haptic stimuli to reduce fidgeting.

The wrist was chosen because it can provide vibrations without the need for a handheld device, which can be used while doing tasks like typing or writing. Another benefit of using the wrist is that accessories currently exist for the wrist such as watches and jewelry. So, a device designed to be worn on the wrist will blend in for any public or professional setting. This paper will explore if haptic fidgeting can affect concentration and performance on a low cognitively-demanding task, which is designed to induce fidgeting. Participants will be observed throughout the test to track how frequent they fidget. The test will also provide performance metrics such as accuracy and reaction time. Lastly, the qualitative data will be gathered by asking participants a series of questions to measure how well they liked the device.

This can be used in everyday life such as driving, meditation, or work and school environments and will be especially be useful for those with attention disorders, such as Attention-Deficit Disorder (ADD) and Attention-Deficit/Hyperactivity Disorder (ADHD). While the vibration patterns tested in this paper were aimed to limit fidgeting and mind wandering, the device can be extended to more applications. This is discussed in the future work section in more detail.

II. RELATED WORK

There has been significant research in the application of vibration stimuli to suppress a variety of unwanted effects. This has been studied for rehabilitation techniques for many different disabilities. For example, a recent study found by applying an imperceivable vibration to the wrist of stroke patients, tactile sensation at the finger pads can improve tactile sensation that was lost from the stroke [9]. Additionally, researchers have explored the effect vibration can have on mitigating involuntary movements, such as tremors experienced by patients with Parkinson's Disease (PD). Different frequencies and amplitudes of vibration stimuli were studied to find the optimal vibration stimulus for the specific goal of improving gait disturbances caused by PD [8].

Research is beginning to implement vibration to prevent a wide range of undesirable effects beyond rehabilitation. There have been many attempts at utilizing vibration to

prevent motion sickness through acceleration cues [7] and through a constant bone conduction vibration stimulus [6]. Vibration has also been used as a non-invasive method to improve sleep in those with sleeping disorders [11]. However, there is limited research in mitigating the tendency to fidget and maintain attention, despite being a widespread issue.

Fidget Widgets is a device that uses a display screen to provide visual feedback in response to interactions between the user and the device in various configurations [1]. For example, one configuration shows a single bubble from a bubble wrap sheet which can be popped or inflated. This uses the known method of fidgeting by popping bubble wrap to influence the design of the device to allow for endless bubble-wrap popping. To help the user focus, instead of mitigating fidgeting, this device simply makes the fidgeting more convenient and accessible.

Similarly, Williams et. al examined a haptic fidget interaction between the user and a mobile device. They found that tactile cues could provide a calming effect during tasks requiring the user's attention [3]. However, the device still required the user to interact with a mobile device, which is not possible while completing tasks requiring both hands, such as typing.

Overall, prior works indicate a growing interest in methods of vibrotactile feedback in rehabilitation, improving attention, and task performance, but lack a solution that leaves both hands unencumbered. We plan to continue this, with a wearable device that intentionally mimics natural fidgeting behaviors by applying similar stimuli at the wrist through vibration. This allows the device to be used for mitigating attention loss for a broader range of task applications.

III. Design

The Vibrotactile Fidget Band, in comparison to the technology stated previously, is a wearable device that administers vibrations patterns to minimize mind-wandering and fidgeting tendencies. The design takes the need for stimulation away from the user's hands and directs it to the user's wrist, thus limiting the desire for the wearer to stimulate through other means. The main device is implemented into a wristband casing as shown in Figure [1].

Figure [2]: Vibrotactile Fidget Band casing.

The device employs an Arduino Nano for the microcontroller, implementing code containing vibrational patterns, a 3.7-Volt Lithium-ion battery, and six flat coin button-type vibration motors. With the electronics and the wristband, the device stayed within our $50 budget. All motors are attached to the wristband, where two motors will be located corresponding to the top of the wrist and three are located on the bottom of the wrist, when the band is worn. This is a total of 5 vibration motors, which translates well to fidgeting commonly done with the fingers because each finger will correspond to the motor, making the vibration patterns feel more natural and intuitive The bottom of the wrist has an additional motor as opposed to the top, because the palm side of the wrist has superior tactile acuity.

The completely assembled device is shown in Figure [3]. Future iterations of the device will aim to limit the size and comfort when worn around the wrist.

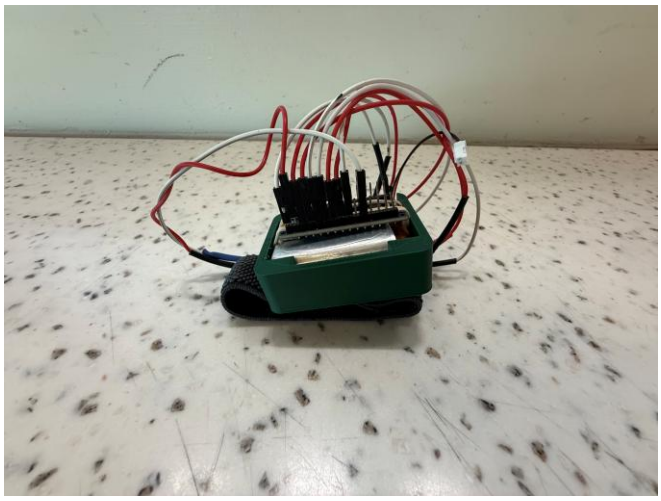

Figure [3]: Fully assemble vibrotactile fidget band.

IV. Experimental Procedure

For experimentation, willing participants will be administered a low-cognitive test to induce fidgeting and mind-wandering (MW). This test, namely the PEBL Continuous Performance Test, will display singular letters at random in a flashing pattern, with varying timing. The participant is directed to press spacebar for every letter that is shown except for the letter "X", where they are directed to do nothing until the next letter appears. This task induces a low cognitively-demanding state, with the intention to induce fidgeting and mind wandering. The test lasts for 14 minutes, and will display the total amount of hits, misses (omissions), false alarms (commissions), and the user's reaction time.

The testing environment will include a computer for taking the PEBL CPT, other common school supplies users might employ to fidget, i.e. pencils, pens, and paper clips, and a camera to record participants while they are completing the test. This is for the experimenter to record and review the amount of fidgeting the participant does while completing the test. For this task, fidgeting is defined as the following:

- Object manipulation that is not required for the task, such as:
  - Clicking pens
  - Twirling pencils
  - Bending paper clips
  - Tapping objects on the desk
  - Sliding items back and forth
- Body fidgeting:
  - Leg bouncing
  - Foot tapping
  - Finger drumming
  - Repeated shifting in seat
  - Restless movement
  - Nail picking
  - Repeated touching of the face
  - Hair twirling
  - Continuous rubbing of hands and arms

All these tendencies can be recorded within the experiment to be deemed as fidgeting. Isolated adjustments, such as scratching and repositioning, or task-related movement will not be counted toward fidgeting.

The following hypotheses are to be tested:

[H1] The user will record improved accuracy and reaction time in the PEBL CPT when using the Vibrotactile Fidget Band compared to when the device is unemployed.

[H2] The user will fidget less when the Vibrotactile Fidget Band is used.

[H3] The user will prefer using the Vibrotactile Fidget Band when taking cognitive tests.

The order of testing is as follows:

1. Participants will be sat acknowledging their willingness to participate in an experiment. The participants will not be informed this experiment will be tracking their fidgeting to prevent any bias.
2. Participants will complete the PEBL CPT without employing the vibrotactile fidget band as a control group.
3. Participants will be asked to evaluate their concentration during the test, on a 0-10 scale, where 0 is not paying attention at all and 10 is complete concentration.
4. The participants will be given the vibrotactile fidget band to wear and steps 2 and 3 will be repeated.
5. Participants will be asked to evaluate their concentration during the test, ranking on the same 0-10 scale.
6. Participants will be asked if they preferred taking the test with or without the vibrotactile fidget band.

Upon the completion of the task, the baseline and experimental data will be pulled from the PEBL CPT and converted to percentages, weighing the ratio of hits to misses. Real time data will also be compared. Along with testing data, video data recording the participant will be diagnosed, converting fidgeting tendencies into statistical data. The baseline recording will then be compared to the experimental recording and charted accordingly. Finally, the user response to the concentration rating scale will be compared, as well as the survey inquiring the user's preference.

V. Results

The experiment yielded 3 quantitative measurements: Accuracy in table 1, reaction time in table 2, and fidgeting frequency in table 3. The accuracy is a combination of both omissions and commissions and converted into a percentage. The reaction time is the time taken to press space bar after a letter that is not "X" appears, measured in milliseconds. The fidgeting frequency was a measure of fidgeting occurrences of the 14-minute testing period.

Accuracy (%)		
Participant	With Vibrotactile Fidget Band	Without Vibrotactile Fidget Band
1	96	98
2	95	99
3	99	98
4	90	88
5	95	95
6	85	86
7	87	85
8	85	88
9	86	81
10	95	85

Table [1]: PEBL CPT accuracy data.

Reaction Time (milliseconds)		
Participant	With Vibrotactile Fidget Band	Without Vibrotactile Fidget Band
1	428.11	437.00
2	424.54	453.33
3	465.63	436.47
4	375.52	363.17
5	367.38	367.05
6	347.49	328.08
7	342.55	325.76
8	343.41	334.86
9	332.88	336.24
10	334.75	349.21

Table [2]: PEBL CPT reaction time data.

Fidgeting Frequency		
Participant	With Vibrotactile Fidget Band	Without Vibrotactile Fidget Band
1	6	43
2	18	27
3	20	38
4	29	36
5	9	22
6	3	21
7	9	14
8	10	19
9	14	18
10	4	11

Table [3]: Number of observed fidgeting occurrences throughout the 14-minute PEBL CPT test.

From the data gathered, we have concluded that the vibrotactile fidget band reduced the fidgeting frequency of the participants with statistical significance, using a repeated measures ANOVA test, shown in figure [4].

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Band	Sphericity Assumed	806.450	1	806.450	16.607	.003
	Greenhouse-Geisser	806.450	1.000	806.450	16.607	.003
	Huynh-Feldt	806.450	1.000	806.450	16.607	.003
	Lower-bound	806.450	1.000	806.450	16.607	.003
Error(Band)	Sphericity Assumed	437.050	9	48.561		
	Greenhouse-Geisser	437.050	9.000	48.561		
	Huynh-Feldt	437.050	9.000	48.561		
	Lower-bound	437.050	9.000	48.561		

Figure [4]: Repeated Measures ANOVA test results for fidgeting frequency.

Additionally, the PEBL test accuracy and reaction times were recorded and analyzed. However, these did not show any significant difference between wearing the vibrotactile fidget band and without. We hypothesize that this could be due to the test being very simple, so the difference in performance will be marginal. Furthermore, vibration has been known to produce a calming effect, which we observed during these trials and this calming effect could actually hinder performance metrics such as reaction.

VI. Discussion

The testing of the Vibrotactile Fidget Band led to various results in which need to be analyzed. In relation to the PEBL test's statistical results, there were a few conclusions that had been reached - the first of which is related to reaction time and accuracy. When observing the data, the use of the Vibrotactile Fidget Band did not lead to statistically significant data. This is seen in Table [1] and Table [2], where there is no meaningful difference between when the band is worn versus when it is not. These results conclude that **[H1]** is not statistically supported, therefore **[H1]** is false and is not explored in further testing.

When observing a trend in fidgeting tendency when the fidget band is employed, however, there is a statistically significant difference shown. This is determined by a Repeated Measures ANOVA test, in which the numerically counted fidgets are compared when the Vibrotactile Fidget Band is used and when it is not. The data shown in Table [3] is evident of a consistent trend, in which the Vibrotactile Fidget Band works as intended, confirming the hypothesis set in **[H2]**. This hypothesis states that the user will have a mitigated desire to fidget, thus proving that the fidget band provides adequate stimulus preventing mind wandering. Further testing is desired to consistently prove this hypothesis, however with the adequate amount of subjects and trial time the data is supported with relevance.

In relation to **[H3]**, which states that users will prefer using the Vibrotactile Fidget Band over not, a survey was conducted. The results showed a trend: 8 participants said they preferred the device, 1 said they did not prefer the device, and 1 was indifferent. This shows that overall, users preferred the device and would use it again if provided the opportunity. This confirms the hypothesis and shows that two out of the three proposed theories were confirmed by the innovated device. Furthermore, 8 of the users said they would buy the device if made mainstream.

Alongside these results, at the end of the test each participant was asked for suggested work. Among the responses included the following:

- Varying and modifiable intensity
- Interchangeable stimulation presets
- Randomized stimulation

These suggestions will be implemented into future iterations of the device and will be provided in a mainstream generation of the fidget band.

During the conduction of the tests, there was an interesting trend that developed. Among the ten participants, three of them were diagnosed with ADHD. Of the three participants, each of them showed a pattern. When the wristband was employed, the participants were noticeably calmer and more concentrated, reporting that there was noticeable peace among them when conducting the PEBL test with the band as opposed to without. Although this concentration did not improve PEBL test reaction time and accuracy results, it was a noteworthy effect that calmed the user. This trend was also noticeable in observation, as users were less extreme in reaction to mistakes. This was not necessarily a recordable statistical result, however, was worthy enough of a mention and possible future test integration.

VII. Future work and Limitations

While this device did significantly reduce fidgeting among our participants, there were significant drawbacks. To start, some common methods of fidgeting are difficult to translate into vibrational patterns. The eccentric rotating mass vibration motors used in this device take time to begin and finish spinning. This introduces a significant latency that limits the ability to provide sharp pulses and stimuli. This hinders the realism of the patterns, as most fidgets are crisp stimuli. To improve this, future iterations should consider using alternative actuators such as linear resonant actuators capable of providing a sharp stimulus with minimal latency introduced into the system.

The current design uses a flexible band that can stretch to fit different wrist sizes. However, there was no way to make the band tighter to fit smaller wrists. The next iteration should consider a silicon band with a clasp mechanism similar to those commonly used in smart watches. This will allow for the device to fit on a larger range of wrist sizes. In addition, this design secured the motors to the band using a strong glue specialized for fabric. This was done to fit the initial budget, but there was an issue with the glue not holding one of the motors, as well as one of the motor leads being torn off after repeatedly taking the device on and off. There are more robust methods to secure and protect the motors that should be considered.

During the experimentation, the device stayed plugged into the computer that was running the PEBL CPT test, via a USB to USB-C. This primarily was done so the participant could upload all vibration patterns quickly to find their preferred pattern. To allow for the user to have full mobility, the device should be completely wireless. So, a feature that allows for toggling between patterns on the device itself would greatly improve the experience. This could be done touch screen, display screen with a toggle button, or a smartphone app that connects to the fidget band. Additionally, this would allow for adding a way to customize the frequency and amplitude of the vibration to each user's preferences.

The biggest improvement that could be made is reducing the size. The housing for the Arduino Nano and battery was the length and width of a typical watch as intended. However, it was much thicker than any typical wrist-worn accessory. This made the device look and feel too bulky and unnatural. The Arduino Nano was designed to fit a large variety of applications. So, although it is the smallest Arduino, it is bulky because of its generalized design. By replacing the Arduino and developing a custom PCB and MPU, the size can be greatly reduced. Furthermore, using a smaller battery will help as well. Although, these changes will increase the overall cost of the device.

As mentioned in the related work section, vibrations have been used to accomplish many different tasks other than fidgeting. In fact, there are current products that help users with mental health — such as anxiety and depression — menstrual health, but there is not much scientific research published for these applications yet. In the future, this device could test and implement vibrations for these applications and those mentioned in the previous work section. Additionally, to improve its performance, sensors could be added to create a closed loop system. There is current research being done to detect and predict when fidgeting and mind wandering is occurring, which could be implemented into this device [10]. Similarly, if the device was expanded into other applications, technology to track sleep stages, anxiety, etc. currently exist and are widely used, which could be integrated into this device, if it is expanded into those applications.

VIII. Conclusion

To conclude, the Vibrotactile Fidget Band yielded pleasant results relating to the hypotheses tested. Although experimental results did not support PEBL test accuracy and reaction time, it properly mitigated fidget tendencies and showed room for further work and implementation. Not only this, but it also showed to be preferable to users as opposed to instances where it was not employed.

The fidget band also showed interesting results in relation to those diagnosed with ADHD and hyperactivity, which shows the possibility for further work in exposure therapy. The next steps into the future of this device include implementing new vibrational patterns and intensity, allowing for user dexterity. Upon innovation, the device and idea will be set forth into patent development and furthered into mainstream technological advancement.

Overall, the device works as intended for which it was designed and will be tested and designed further for more various implementation scenarios. The conclusion of the experimental results proved to be statistically significant and therefore this product is statistically supported as properly operational in nature.

APPENDIX

REFERENCES

[1] M. Karlesky and K. Isbister, "Fidget Widgets: Secondary Playful Interactions in Support of Primary Serious Tasks," in *CHI '13 Extended Abstracts on Human Factors in Computing Systems*, 2013, pp. 1149–1154.

[2] Eichenlaub, J. A., Huisman, G., & Xue, H. (Year). *Exploring extended mind-wandering through an interactive haptic fidget object*. Delft University of Technology.

[3] Williams, R. M., Posadas, B., Prioleau, D., Laurenceau, I., & Gilbert, J. E. (2019). *User perceptions of haptic fidgets on mobile devices for attention and task performance*.

[4] Bruns Alonso, M., Hummels, C.C.M., Keyson, D.V. *et al*. Measuring and adapting behavior during product interaction to influence affect. *Pers Ubiquit Comput* **17**, 81–91 (2013).

[5] Choi, W., Song, A., Edge, D., Fukumoto, M., Lee, U.: Exploring user experiences of active workstations. In: Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing – UbiComp 2016. pp. 805–816. ACM Press, New York, USA (2016)

[6] S. Salter *et al*., "Evaluation of Bone Conducted Vibration as a Potential Countermeasure to Motion Sickness in Automated Vehicles," Coventry University and Jaguar Land Rover, UK, and Otolith Labs / University of Maryland School of Medicine, USA.

[7] W. Emond and M. Zare, "*Can acceleration-coded vibrations help alleviate passenger motion sickness? Investigation in stop-and-go passenger rides*," *Applied Ergonomics*, vol. 132, p. 104682, 2025, doi: 10.1016/j.apergo.2025.104682.

[8] I. Pretzer-Aboff, R. K. Elswick, A. Gouelle, N. Helm, G. Blackwell, and L. Cloud, "*Determination of optimal vibration dose to treat Parkinson's disease gait symptoms: A clinical trial*," *Clinical Parkinsonism & Related Disorders*, vol. 10, p. 100248, 2024, doi: 10.1016/j.prdoa.2024.100248.

[9] Lakshminarayanan K, Lauer AW, Ramakrishnan V, Webster JG, Seo NJ. Application of vibration to wrist and hand skin affects fingertip tactile sensation. Physiol Rep. 2015 Jul 14;3(7):e12465. doi: 10.14814/phy2.12465. Epub 2015 Jul 14. PMID: 26177959; PMCID: PMC4552540.

[10] B. Bühler, E. Bozkir, H. Deininger, P. Goldberg, P. Gerjets, U. Trautwein, and E. Kasneci, "*Detecting aware and unaware mind wandering during lecture viewing: A multimodal machine learning approach using eye tracking, facial videos and physiological data*," in Proc. 26th Int. Conf. Multimodal Interaction (ICMI '24), 2024, pp. 244–253, doi: 10.1145/3678957.3685710.

[11] H. B. Kwon, J. Jeong, B. Choi, K. S. Park, E. Y. Joo, and H. Yoon, "*Effect of closed-loop vibration stimulation on sleep quality for poor sleepers*," *Frontiers in Neuroscience*, vol. 18, Art. no. 1456237, 2024, doi: 10.3389/fnins.2024.1456237.